P-RANGER FLIGHT CONTROLLER

V1.0 Mini

OFFICIAL TECHNICAL & USER MANUAL

Installation, Configuration, PID Tuning, and UAV Integration Guide

Designed & Developed by:

Muhammad Waseem

Developer, P-RANGER Flight Systems

Proudly Made in Pakistan

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Author's Note: The P-RANGER Journey

The idea for the P-RANGER flight controller was born out of a simple vision: to design a reliable drone controller using microcontrollers and sensors that are easily available in Pakistan. I wanted to create an affordable, highly accessible platform within the reach of students, learners, and hobbyists.

Developing indigenous hardware comes with unique challenges. When I contacted local PCB manufacturing companies, I discovered their fabrication limitations—they could not produce boards with more than two layers, nor did they support drill hole sizes smaller than 0.5mm. I adapted to these constraints, completely restricting and optimizing my design strictly to a 2-layer stackup to ensure local manufacturability.

When the bare PCBs finally arrived, I hand-soldered every single component myself. I conducted the initial power-on and sensor self-tests, which all passed perfectly. After compiling and loading the custom firmware, overcoming numerous technical hardships along the way, the flight controller was finally ready for the skies.

This project required immense effort and was funded entirely from my personal, hard-earned savings—a significant financial challenge for me. P-RANGER is an excellent educational tool for anyone looking to learn manual PID tuning, understand basic drone mechanics, and master RC flight. Above all, it is proudly "Made in Pakistan." I hope this platform helps others learn, and I humbly request your support as I continue developing local hardware.

Muhammad Waseem

Developer & Hardware Designer, P-RANGER Flight Systems

Made in Pakistan

P-RANGER Flight Controller (V1.0) – Technical Manual

This complete installation, configuration, and tuning manual applies to the P-RANGER Mini flight controller platform (V1.0), designed for custom airborne equipment development, UAV research, and system integration.

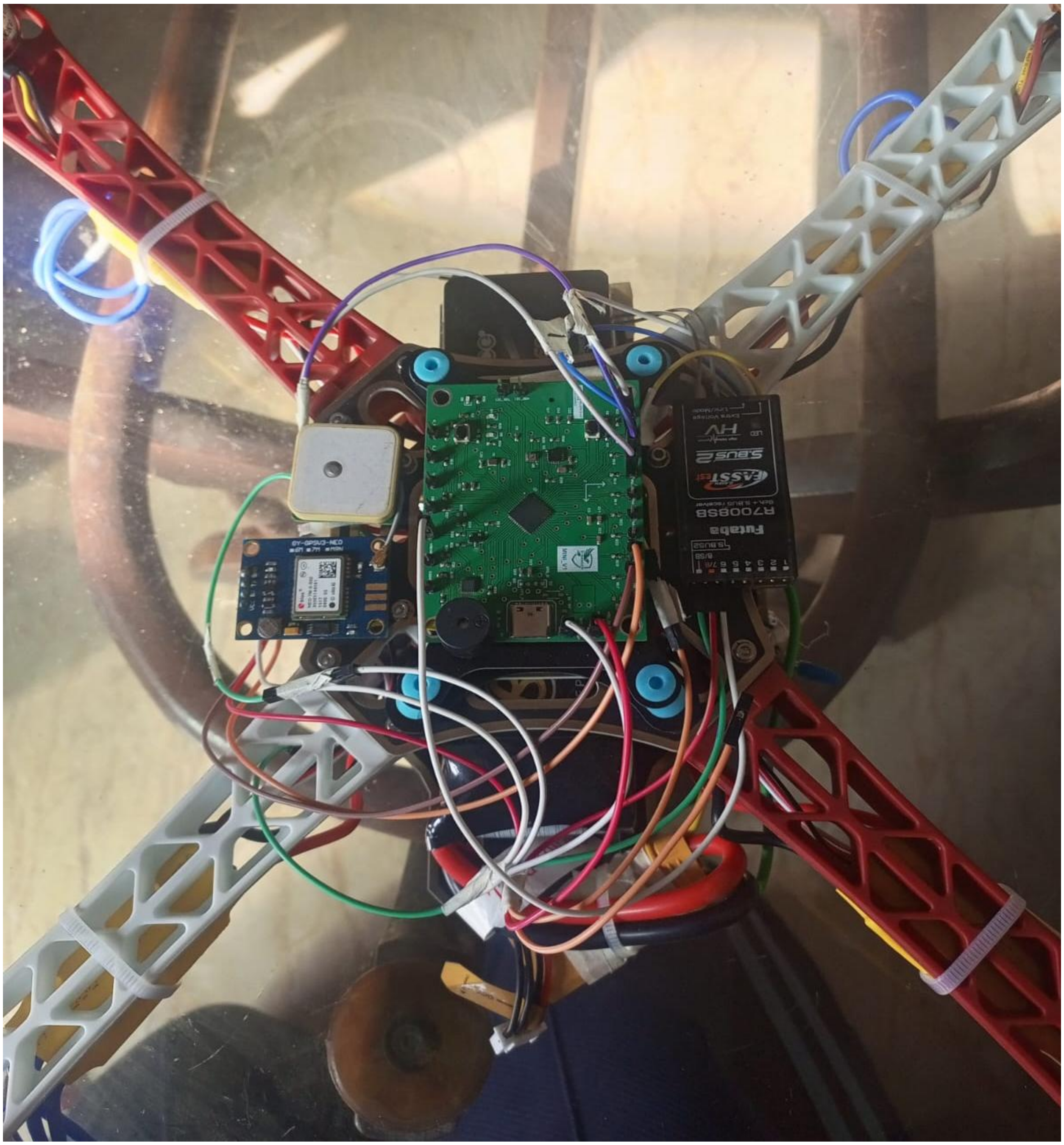

Figure 1: P-RANGER Install on F450 Frame

1. System Overview & Technical Specifications

The P-RANGER is an advanced, high-density flight controller optimized for operation, high stability, and custom firmware target builds.

Feature	Specification
PCB Dimensions	63 mm × 48.5 mm
Stackup	2-Layer Custom PCB Design
Processor	High-Performance STM32 Series MCU
Primary IMU	MPU6500 upgrade able to MPU9250 (Gyro/Accelerometer)
Altitude Sensing	BMP280
Compass	QMC5883L
Onboard Storage	16MB SPI Flash IC (W25Q16 Series)
External Storage	MicroSD Card Slot Interface
Firmware Support	INAV & Beta Flight (Primary INAV)

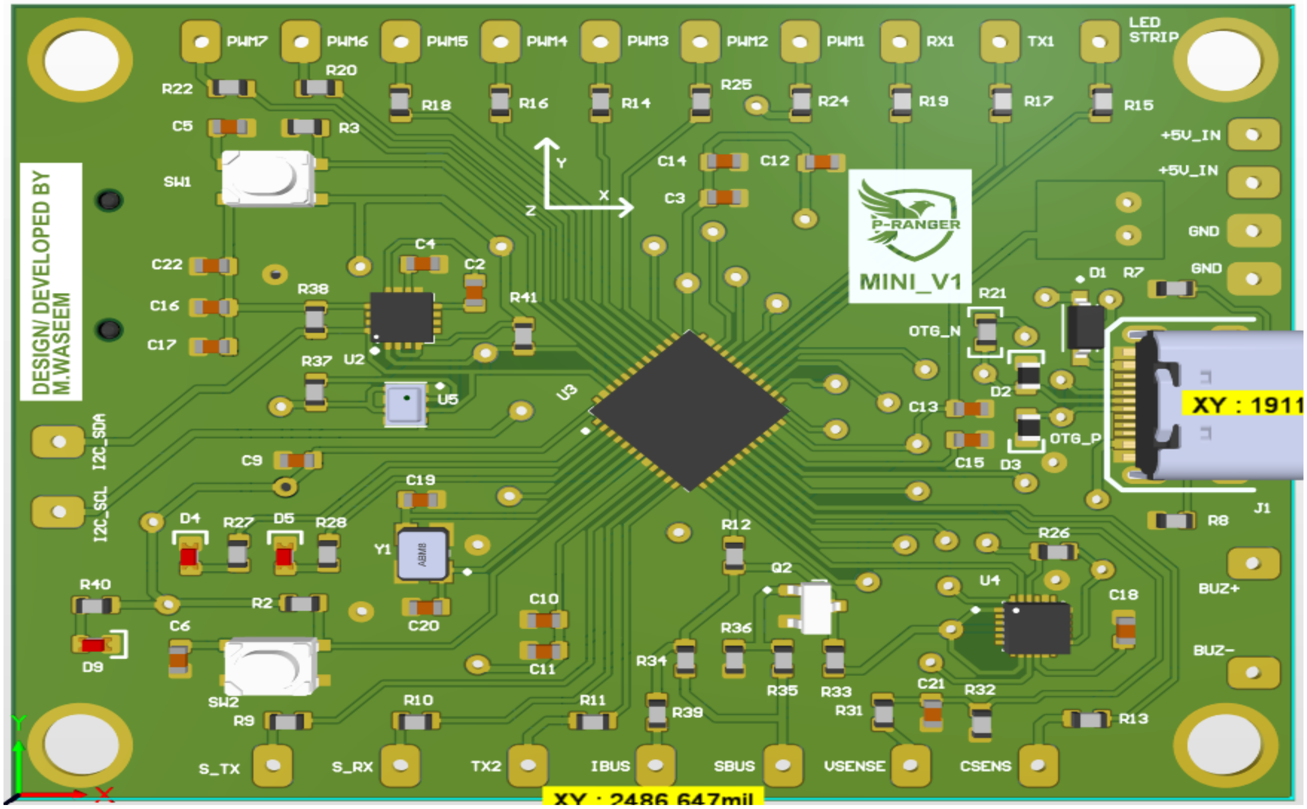

Figure 1: Top view of PCB

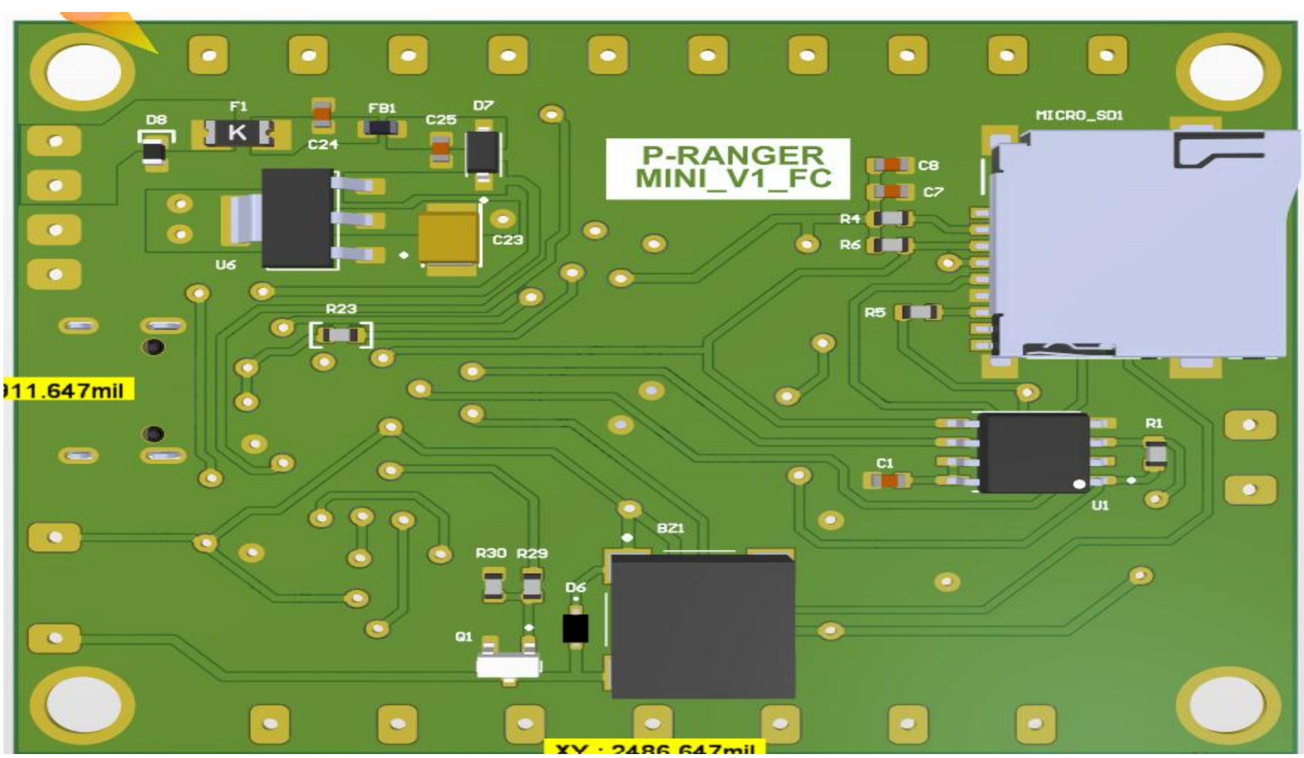

Figure 2: Bottom view of PCB

2. Power Architecture & Voltage/Current Sensing

- **Main Power Input:** Supply exactly **+5V DC** to the designated power pins. The board utilizes an onboard AMS1117 3.3V LDO regulator; strictly adhere to the 5V input limit to prevent thermal overload.
- **Reverse Polarity Protection:** Reverse polarity protection circuitry is integrated into the design. However, reverse connection must be avoided to prevent stress on protection components.
- **USB Port Data/ Power Routing:** The onboard Micro-USB / Type-C connector routes **only the D+ and D- data lines** to the MCU. To connect the flight controller to a PC configurator, you **must supply +5V DC externally** (via a buck converter or BEC to the 5V input pins) to power the board logic.
- **Voltage Sensing (**vsense**):** Connect your battery main supply (up to 3S) directly to the vsense pin. In the INAV Battery tab, adjust the voltage scale setting until the reported voltage matches a digital multimeter measurement.
- **Current Sensing (**csense**):** Interface an external current sensor module output to the csense pin. Ensure the analog signal fed to this pin **never exceeds 3.3V**.

3. Mechanical Installation & Sensor Isolation

Proper mechanical isolation is critical for accurate state estimation and preventing flyaway conditions on multirotor frames (such as F450 class builds).

- **Vibration Damping: Never hard-mount the FC board directly to the frame using rigid metal screws.** High mechanical vibration levels (accVib) degrade accelerometer performance. Mount the board using rubber anti-vibration standoffs (bobbins) or multi-layered double-sided foam tape.
- **Barometer Shielding:** Place dark, open-cell foam tightly over the barometer sensor. Unshielded barometers are susceptible to propeller draft (prop-wash) and sudden ambient light changes, which cause false altitude readings, sudden flips, or erratic vertical climbs during NAV ALTHOLD.
- **GPS/Compass Placement:** Mount the external GPS/Magnetometer module on an elevated mast at least 8–10 cm above the frame to isolate the compass sensor from electromagnetic interference (EMI) generated by high-current battery wires and ESCs.

4. Hardware Interfaces & Port Assignments

Interface / Pin Label	**Primary Function**	**Configuration Notes**
PWM Outputs 1–7	Servo / ESC Control Signals	7 dedicated PWM output pins clearly labeled for motor and actuator control.
I2C Interface	Dedicated SDA & SCL	External I2C bus for external compass (e.g., QMC5883L) or sensor expansion.
UART 1	GPS Module	Assign **GPS** under Sensor Input in the Ports tab. Set baud rate to 57600 or 115200.
UART 2 (RX)	Serial Receiver Input	Connect SBUS or IBUS signal wire. Enable **Serial RX** on UART 2. *Do not connect SBUS and IBUS simultaneously.*
UART 3	SoftSerial / Custom	SoftSerial port. Configure for auxiliary telemetry or smart audio in the Ports tab as needed.
LED Strip Pin	Dedicated Addressable LED Output	Configured PWM output for WS2812 LED strips. Supply external +5V power to the LEDs and connect only the signal wire to this FC pin.
Buzzer Output	Active Buzzer Driver	Connect a 3.3V active piezo buzzer to the dedicated buzzer pads.

5. Firmware Targets & Build Setup

The P-RANGER platform supports custom **INAV** target compilation.

- Firmware will be available GITHUB under this link,

  https://github.com/wasim9190/P-RANGER-Docs

The flight controller supplied to you comes pre-flashed with the custom P-RANGER V1 firmware right out of the box. However, in the rare event of a firmware crash or if a system recovery becomes necessary, please follow the firmware flashing procedure outlined below.

- **Enter DFU Mode:** Hold down the **BOOT** and NRST button on the flight controller while plugging in the USB cable (with +5V power supplied) and leave NRST button, until INAV Configurator displays **DFU** in the top-right port dropdown.

- **Load Target:** Navigate to the **Firmware Flasher** tab and click **Load Firmware [Local]** to select your custom-built .hex file (or select your target and version from the online list).

- Flash **Board:** Click **Flash Firmware** and wait for the verification process to complete 100% before disconnecting or rebooting the board.

6. Initial INAV Setup & Connection (Post-Firmware Flash)

After successfully flashing the firmware, follow these steps to establish a connection and apply the foundational settings for your airframe:

- **Connect to Configurator:**

  Open the INAV Configurator on your PC.

  Select the correct **COM Port** for your flight controller from the top-right dropdown menu.

  Ensure the Baud Rate is set to **115200**.

  Click the **Connect** button.

- **Apply Mixer & Frame Preset:**

  Navigate to the **Mixer** tab on the left menu.

  Select your vehicle type from the platform options (e.g., Multirotor, Airplane, Rover, or Boat).

Choose the preset that closest matches your frame size (e.g., **Quad X 3", 5", 7", 9"**, or standard Quad X for heavy custom builds like the F450).

Click **Load and Apply** to automatically configure the motor geometry, default rates, and baseline PID profile for your selected size.

Click **Save and Reboot**.

- **Board Alignment & Sensor Setup:**

  Go to the **STATUS** tab.

  Verify that the 3D model on the screen moves exactly as you physically move the drone. If the P-RANGER board arrow is not pointing directly forward on the physical frame, adjust the **Board Alignment** (Yaw degrees) accordingly.

- **Assign Ports:**

  Go to the **Ports** tab.

  Enable **Serial RX** on UART 2 (for your SBUS/IBUS receiver).

  Select **GPS** under Sensor Input for UART 1.

  Switch ON MSP button if telemetry is connected.

  Click **Save and Reboot**.

- **Configure Receiver Protocol:**

  Go to the **Receiver** tab.

  Select **Serial-based receiver** in the telemetry dropdown and choose your specific protocol (**SBUS** or **IBUS**).

  Turn on your RC transmitter and move the sticks to verify that the roll, pitch, yaw, and throttle channels respond correctly in the software. Click **Save**.

7. Flight Modes Configuration

Assigning flight modes allows you to control the aircraft's behavior and autonomous features using the auxiliary (AUX) switches on your RC transmitter. Navigate to the **Modes** tab in the INAV Configurator to set these up:

1. **Adding a Mode:** Click **Add Range** next to the desired mode, select the corresponding AUX channel from the dropdown, and drag the yellow slider to cover the area where your physical switch is active. Click **Save** when finished.

2. **Essential Modes Breakdown:**
   - **ARM:** Enables the motors for flight. Assign this to a dedicated, easily accessible 2-position switch. The flight controller will only arm if the throttle is at absolute zero and all safety checks (like GPS lock) are passed.
   - **ACRO (Rate Mode):** This is the default flight mode in INAV. If no other stabilization modes (like ANGLE) are highlighted in the Modes tab, the drone is flying in ACRO. It does not auto-level and requires manual stick inputs to maintain attitude.
   - **ANGLE:** The primary auto-leveling mode. When you release the pitch and roll sticks, the drone automatically returns to a perfectly flat, level hover. Highly recommended for takeoff, landing, and standard line-of-sight flying.
   - **NAV ALTHOLD:** Utilizes the onboard barometer to maintain the current altitude automatically while you control directional movement. *Safety Warning:* Only engage NAV ALTHOLD while airborne at a stable mid-stick hover. Engaging it on the ground can cause the drone to shoot upwards unexpectedly due to ground-effect air pressure.
   - **NAV POSHOLD:** Utilizes the GPS, compass, and barometer to lock the drone in a fixed 3D position in the air. *Safety Warning:* Do not engage with fewer than 10 satellites. A weak GPS signal can cause dangerous toilet-bowl oscillations.
   - **NAV RTH (Return to Home):** An autonomous safety mode that takes control of the drone, flies it back to the exact GPS coordinates where it was armed, and automatically lands it.
   - **BEEPER:** Triggers the external buzzer manually. Assign this to a spare switch to easily locate the drone if it lands in tall grass or bushes.

8. External Wireless Telemetry Setup

To monitor live flight data wirelessly on a laptop using external telemetry radios (e.g., SiK Radios or 3DR telemetry modules), follow these strict configuration rules to avoid protocol conflicts:

- **Hardware Connection:** Connect the air-side telemetry radio's TX and RX pins directly to a free **Hardware UART** on the P-RANGER board (TX to RX, and RX to TX). Try to **Never use UART 3 (SoftSerial)** for wireless telemetry; software-emulated ports cannot handle the required high data rates (57600+ baud) and will cause packet loss.

- **Radio Firmware Fix (Crucial for SiK Radios):** Telemetry radios commonly used with Pixhawk systems have "MAVLink Framing" enabled by default in their internal firmware. Because INAV Configurator communicates using the **MSP (MultiWii Serial Protocol)**, MAVLink framing will actively block the connection. Connect the radio to your PC and use a configuration tool (like Mission Planner) to completely **disable MAVLink Framing**.
- **INAV Ports Configuration:** In the INAV Ports tab, locate the hardware UART connected to your telemetry module. Enable the **MSP** toggle switch and set the baud rate to **57600** (or match your specific radio's configuration). *Do not select MAVLink or any other protocol under the Telemetry Output dropdown.*
- **Ground Station Connection:** Connect the ground-side telemetry radio to your laptop (via direct USB or a USB-to-TTL converter). Open INAV Configurator, select the converter's COM port, set the connection baud rate to 57600, and click Connect to establish the live wireless data link.

9. Sensor Calibration Procedures

Perform full sensor calibration on a stable, flat surface before initial flight testing.

Accelerometer Calibration (6-Point)

1. Place the drone on a completely level surface.
2. Open the **Setup** tab in INAV Configurator and select **Calibrate Accelerometer**.
3. Complete the 6-position step sequence by resting the aircraft flat on each side (Level, Inverted, Left, Right, Nose Up, Nose Down) when prompted, keeping the board perfectly stationary during measurement capture.

Compass (QMC5883L / External Magnetometer) Calibration

1. Move the aircraft away from large metal structures, power lines, and magnetic sources.
2. Click **Calibrate Magnetometer** in the Configurator.
3. Within the 30-second countdown window, rotate the drone through 360 degrees across all three physical axes (Roll, Pitch, and Yaw) in a continuous smooth motion.
4. Verify that the virtual model heading correctly reflects physical magnetic directions (North, South, East, West).

10. Data Logging & Blackbox Setup

The flight controller includes **16MB onboard SPI Flash (W25Q16)** and a **MicroSD Card slot** sharing the main SPI bus line.

SD Card Partitioning Requirement

If the SD card reports a NO_FAT_MBR_PARTITION error in the CLI, the card lacks an MBR partition table. Standard Windows formatting tools often fail to create an MBR header. Re-partition via Windows Command Prompt (diskpart):

Plaintext →diskpart → list disk → select disk X (Replace X with your SD card disk number)

→ clean → convert mbr → create partition primary → format fs=fat32 quick → active → exit

Blackbox Device Selection

By default, the flight controller routes logging to internal SPI flash. To switch between storage mediums, enter the INAV CLI:

- **Route to SD Card:**

Plaintext

set blackbox_device = SDCARD

save

- **Route to Internal Flash:**

Plaintext

set blackbox_device = SPIFLASH

save

11. Motor & ESC Configuration

1. **Safety Rule:** Remove all propellers before performing motor mapping or ESC calibration.
2. Navigate to the **Outputs** tab in INAV Configurator.
3. Toggle the **"Enable motor and servo output"** safety switch at the bottom of the screen.
4. Connect ESC signal lines to PWM pins 1 through 4 sequentially.
5. Calibrate ESC endpoints (if using standard PWM/Multishot protocols).
6. Test individual motor sliders to confirm that motor numbers (1, 2, 3, and 4) and rotational directions (CW vs. CCW) accurately match the standard quadcopter configuration layout diagram.

12. GPS & Arming Safety Conditions

1. Go to the Configuration / GPS tab and enable the **GPS** toggle switch. Select protocol **UBLOX**.
2. **Arming Safety Requirements:** The flight controller enforces a mandatory GPS lock condition by default.
3. The board will prevent system arming until a minimum lock of **6 GPS satellites** is achieved.
4. For autonomous modes (NAV POSHOLD / NAV ALTHOLD), wait for **10 to 12 satellites** and an **HDOP < 2.0** to prevent position drift or toilet-bowl oscillations.

13. Heavy Airframe PID Tuning

Flight Mode Deployment Strategy

- **ANGLE Mode:** Primary take-off and landing mode. Provides auto-leveling with manual throttle control.
- **NAV ALTHOLD:** Engage **only while airborne at a stable hover height**. Avoid engaging ALTHOLD while resting on the ground; ground-effect air turbulence near the barometer can cause the controller to apply full throttle abruptly upon spin-up.
- **NAV POSHOLD:** Engage only after confirming a strong 3D GPS fix (10+ satellites). Do not engage on marginal fixes.
- **You can use other modes as per your requirements.**

Baseline PID Tuning (Frame F450 / 8045 propellers)

Large airframes operating high-inertia propellers (e.g., 10-inch) require distinct PID adjustments compared to high-KV racing quads:

- **P-to-D Ratio Rule:** Keep the **D-term value significantly lower than the P-term value** (typically around half the P-term). An excessively high D-term relative to P causes sluggish stick feel, delayed response, and motor overheating.
- **Suppressing I-Term Wobble:** High mechanical frame vibrations (accVib > 2000) cause the I-term accumulator to over-correct, generating slow periodic oscillations (wobbling). Maintain low I-term values (e.g., 10–15) until physical vibration isolation (bobbins and prop balancing) is resolved.

Recommended Baseline Starting Parameters (F450 Class)

Roll: P = 40 | I = 15 | D = 18 | FF = 30

Pitch: P = 40 | I = 15 | D = 18 | FF = 30

Yaw: P = 55 | I = 45 | D = 0 | FF = 75

Adjust parameters incrementally in steps of ±3 based on flight-log evaluation.

14. Conclusion & End Notes

The development of the P-RANGER V1 and Mini flight controllers is a testament to the potential of indigenous technology and dedicated, hands-on research and development. Whether you are a student exploring the fundamentals of drone mechanics, a hobbyist mastering manual PID tuning, or a professional working on airborne system integration, this hardware was designed to empower your journey.

Drawing upon more than two decades of practical avionics experience, circuit reverse engineering, and airborne equipment development, my ultimate vision has been to make highly reliable, locally manufactured platforms accessible to everyone. The sky is not the limit; it is simply the starting point for our innovation.

Thank you for supporting this initiative and choosing a proudly "Made in Pakistan" platform. Fly safe, trust your instruments, and never stop experimenting.

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